



2.2 Financial Planning

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- * Sales Forecasting
- * Sales, Revenue & Costs
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- * Budgets



The purpose of sales forecasts

- Sales forecasts **predict future revenues** based on past sales figures
- They commonly focus on what will happen **in the future** to:
 - The volume and value of sales
 - The size of the market
 - Sales as a result of promotional activity
 - Sales as a result of cyclical factors
- Sales forecasts are an important tool to **support planning** and can improve the **validity** of **cash flow forecasts**
- Businesses use sales forecasts to **determine resource requirements** in a variety of ways, including:
 - How many staff will be needed?
 - How much stock will be required?
 - Does **capacity** need to be expanded (or reduced)?
 - Does the equipment need to be upgraded, replaced or increased (or reduced)?
 - How much and **what type of finance** will be required?
 - Is **promotional activity** (e.g. advertising) required — and when?

Factors affecting sales forecasts

- Developing **accurate sales forecasts** is a skill that requires an understanding of several factors that can influence the reliability of the forecast

Consumer trends

- **Seasonal variations**
 - Demand for certain goods is seasonal
 - Events such as **major religious festivals, holiday periods and annual events** impact demand for a wide range of products
 - For example, sales of basic homewares increase when students start university each September
- **Fashion**
 - Fashion is often led by celebrities, and their influence can have a **short-term impact** on sales



- For example, when Hollywood legend Megan Fox appeared in September 2021 at a star-studded event in a Boohoo dress, the company's sales unexpectedly soared by more than 400% during that month

■ Long-term trends

- Consumer behaviour, attitudes and spending habits **change over time**
 - In recent years, environmentally conscious consumers have led many businesses to amend sales forecasts to reflect increased demand for green products
 - For example, in late 2022, vehicle manufacturer Ford increased its sales forecasts for electric vehicles by almost 70%

Economic variables

■ Economic growth

- During periods of **economic growth**, **increased consumer incomes** will lead to higher-than-forecast sales
- The opposite will occur during periods of **economic slowdown**, and sales may be lower than forecast

■ Inflation

- The general increase in prices over time **reduces consumers' spending power**
 - Firms may revise their sales forecasts downwards during periods of rising inflation
 - Firms may revise their sales forecasts upwards during periods of falling inflation

■ Unemployment

- Increased levels of unemployment are often experienced during periods of **recession** and tend to be a key cause of **reduced spending in the economy**
 - Sales forecasts for **lifestyle** and luxury goods may reduce as consumers focus their spending on essentials

■ Interest rates

- When **interest rates** rise, borrowing becomes **more expensive for consumers**
- Businesses that sell products that consumers frequently buy on **credit** may therefore adjust their sales forecasts downwards
 - For example, property sales are set to drop to 1.01bn in 2023 from 1.27bn in 2022, causing many estate agencies to adjust their sales forecasts downwards

■ Exchange rates

- When the value of the UK pound sterling falls against other global currencies, overseas consumers will find that British **exports** become relatively cheaper
- Businesses that sell products overseas or that **cater to tourists** visiting the UK may adjust their sales forecasts upwards to reflect the expected increase in demand

from a cheaper sterling

- For example, Visit Britain expects the number of tourists entering Britain in 2023 to be 14% higher than in 2022



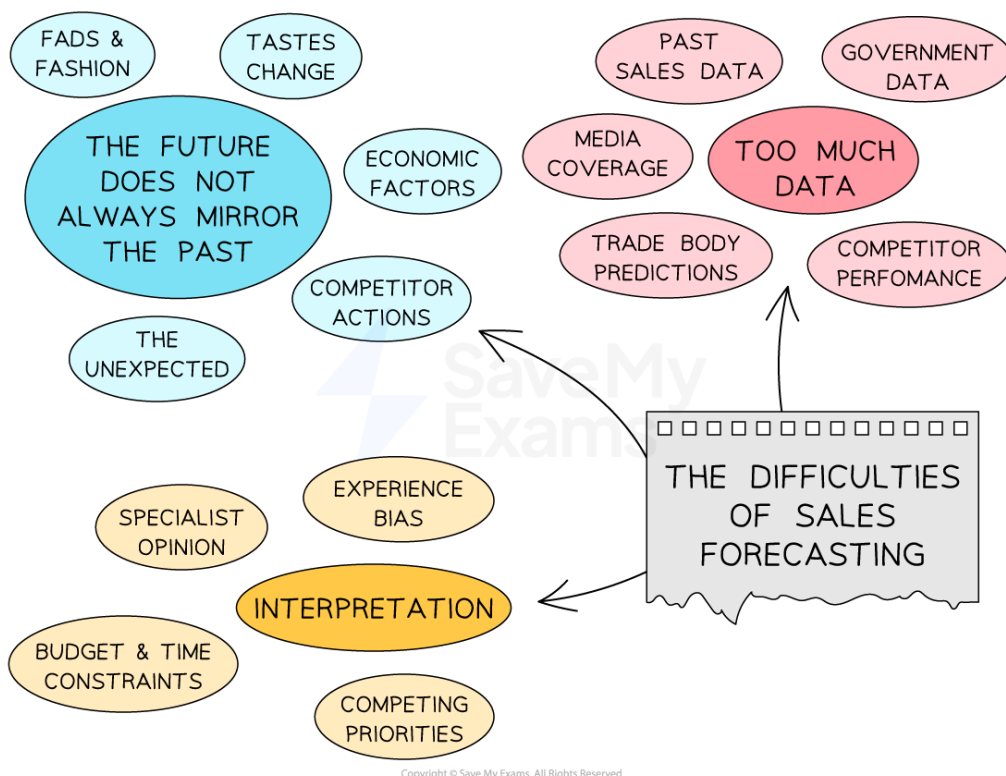
Actions of competitors

- Sales forecasts should consider the **short-term actions of competitors**, such as **sales promotions**, as well as longer-term strategies, such as **changes to product ranges** and expansion plans
- Competitor actions are difficult to predict, so the usefulness of past data to predict future sales may be limited
 - For example, Marks & Spencer announced plans to open 20 new high street stores in 2023, partly in response to the closure of several key competitors, including Debenhams

Difficulties of sales forecasting

- Sales forecasting usually involves **the use of past data** to predict the future
- In the short term, sales forecasts are likely to reflect the recent past
- Longer-term sales forecasting is often more problematic, as **several factors affect its reliability**

The main difficulties of sales forecasting



The difficulties of sales forecasting include interpretation, too much data and the fact that the future does not always mirror the past



Your notes

- Effective sales forecasting requires **skill, time and the accurate use of timely data**
 - Smaller businesses in particular may **lack the experience or specialised personnel** to construct, analyse and interpret sales forecasts
 - It is difficult to avoid **experience bias** (e.g. opinions of the future based on experiences in the past)
 - Businesses may face problems in constructing sales forecasts that ignore the priorities of key **stakeholders**
- The future seldom repeats the occurrences of the past
 - Sales forecasts will rarely reflect the **full range of external influences** that can affect future inflows, such as fashions, trends and the actions of competitors
- There is a significant amount of data available for businesses to consider when constructing sales forecasts
 - Internal data, such as **previous sales figures**, will be a key source of information when constructing forecasts
 - Selecting the most appropriate external data to support sales forecasts is extremely challenging and requires careful evaluation



Examiner Tips and Tricks

While sales forecasting as a planning tool has several potential uses for a business, you should carefully consider how a sales forecast is constructed when evaluating its usefulness.

- Who is responsible for the sales forecast?
- Which data is used in its construction?
- How reliable or accurate are the data sources underpinning the forecast?

You may even conclude that no sales forecasting is better than a poorly constructed, biased attempt!



Calculation of sales volume and sales revenue

- Sales volume is the **number of units** sold by a business
 - For example, the number of Harry Styles album download purchases
- Sales revenue is the **value of the units** sold by a business
 - For example, the revenue earned by Apple Music from sales of music downloads
 - Sales revenue is a **key business performance measure** and must be calculated to identify profit
 - Sales revenue is calculated using the formula:

$$\text{Sales revenue} = \text{Selling price} \times \text{Number of units sold}$$

- Sales revenue **usually increases** as the **sales volume increases**
- When a firm sells one product, it is easy to calculate the sales revenue
 - The more products a firm sells, the harder it is to calculate the sales revenue
 - **Computer systems** make it easier to track sales revenue when multiple products are sold by the business



Worked Example

Fotherhill Organics Limited sold 39,264 packs of its specialist compost to mail-order customers in 2022. The price per pack was £8.75. In addition, it sold 4,280 tonnes to gardening businesses for £123.95 per tonne.

Calculate Fotherhill Organics' sales revenue for 2022.

Step 1: Calculate the sales revenue from sales to mail-order customers

$$\begin{aligned} &= 39,264 \text{ packs} \times \text{£}8.75 \\ &= \text{£}343,560 \end{aligned}$$

Step 2: Calculate the sales revenue from sales to gardening businesses

$$\begin{aligned} &= 4,280 \text{ packs} \times \text{£}123.95 \\ &= \text{£}530,506 \end{aligned}$$

Step 3: Add together the two sales revenue figures

$$= £343,560 + £530,506$$

$$= £874,066$$



Your notes

An introduction to costs

- In preparing goods/services for sale, businesses **incur a range of costs**
- These costs can be broken down into different categories

Fixed costs

- Fixed costs (FC) do not change as the level of output changes
 - These have to be paid whether the output is zero or 5,000
 - For example, building rent, management salaries, insurance and bank loan repayments

Variable costs

- Variable costs (VC) vary directly with the output
 - These increase as output increases and vice versa
 - For example, raw material costs and wages of workers directly involved in production

Total costs

- Total costs (TC) are the sum of FC and total VC

Calculation of fixed, variable and total costs

- Based on the above definitions, we can calculate several different types of costs

Formulas to calculate different types of costs

Type of cost	Formula
TC	= Total FC + total VC
Total VC	= VC per unit × quantity
Average TC (unit cost)	= TC ÷ quantity produced

VC per unit	= Total VC ÷ quantity produced
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Your notes

Example: cost calculations where variable costs = \$60

Output (Q)	FC	TVC = \$60 × Q	TC = TFC + TVC	AVC = $\frac{TVC}{Q}$	AC = $\frac{TC}{Q}$
0	200	-	200	-	-
1	200	60	260	60	260
2	200	120	320	60	160
3	200	180	380	60	126.67
4	200	240	440	60	110
5	200	300	500	60	100
6	200	360	560	60	93.33
7	200	420	620	60	88.57
8	200	480	680	60	85

Variable costs per unit

- VC per unit are calculated by adding together the cost of each component or raw material used to produce the unit



Worked Example

Rosebud Aromas manufactures luxury scented candles. The production of each candle incurs the following VC:

VC	Cost per candle (£)
Wax	0.14
Perfume oil	0.72
Telephone bill	24.32
Glass jar	1.46

Outer packaging	0.33
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Your notes

Calculate the VC in pounds sterling for each candle.

Step 1: Identify the VC in the list

A telephone bill is classified as an FC, so it should not be included in the calculation. [1]

Step 2: Total the VC listed

$$= £0.14 + £0.72 + £1.46 + £0.33$$

$$= £2.65$$

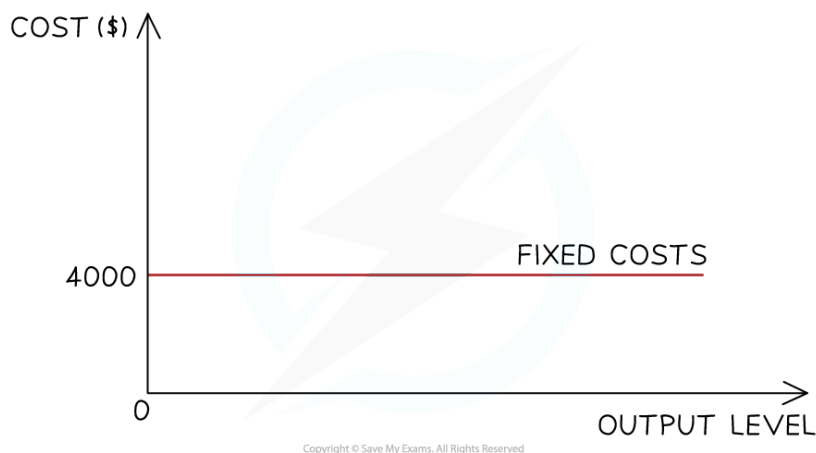


Examiner Tips and Tricks

Take care when calculating VC per unit, as it is likely that one or more **FC** will be included in the list, as seen above.

Diagrammatic representation of costs

Fixed costs



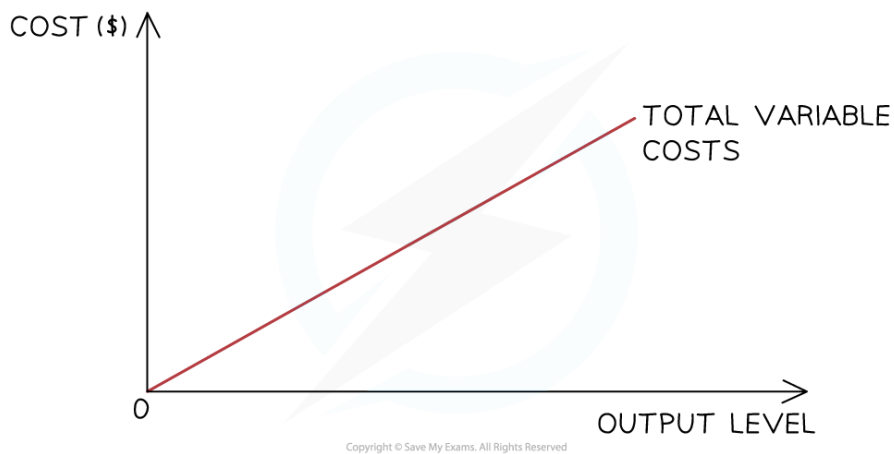
Graph showing fixed costs and output level

- The firm has to pay its **fixed costs**, which do not change according to output
- The FC for this firm are \$4,000

Variable costs



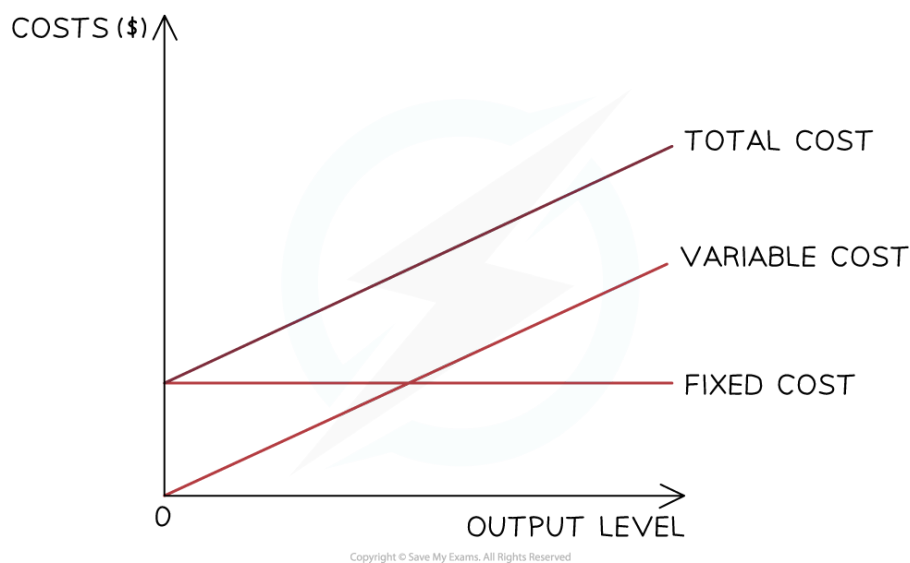
Your notes



Graph showing total variable costs and output level

- The VC initially **rises proportionally with output**, as shown in the diagram
- At some point, the firm will benefit from **purchasing economies of scale**, and the rise will no longer be proportional

Total costs



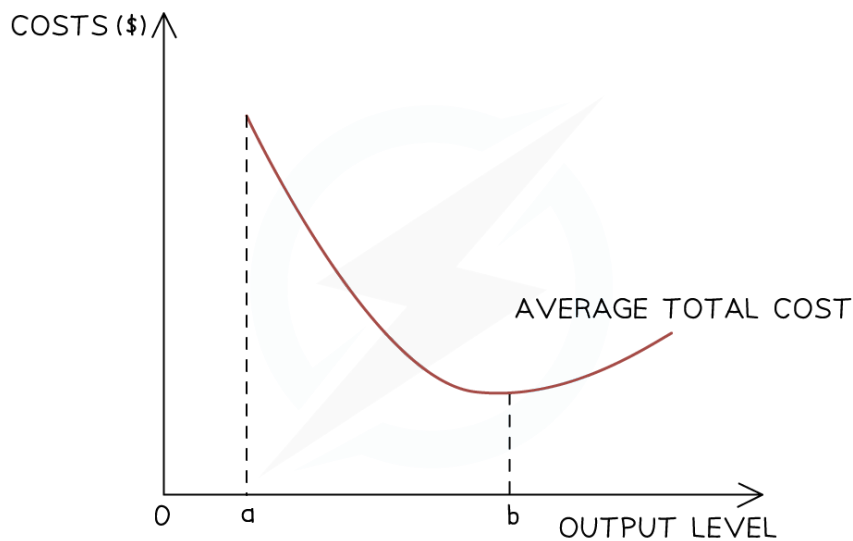
Graph showing a cost analysis involving the fixed cost, variable cost and output level

- The **TC** are the sum of the VC and FC
- The TC **cannot be zero**, as all firms have some level of FC

Average total costs



Your notes



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Graph showing average total cost and output level

- As a firm grows, it can increase its **scale of output**, generating efficiencies that lower its **average total costs (AC)** of production
- These efficiencies are called **economies of scale**
- As a firm continues increasing its **scale of output**, it will reach a point where its **AC** will start to **increase**
- The reasons for the increase in the average costs are called **diseconomies of scale**

Contribution

- Contribution refers to a product's **selling price minus the VC** directly involved in producing that unit
- Contribution can be calculated using the **following formula**:
$$\text{Contribution} = \text{Selling price per unit} - \text{Variable cost per unit}$$
- It is called "contribution", as this amount contributes towards **paying off the FC** of the business
 - Once the FC have been paid off, the contribution starts to **contribute to the profits** of the business
- Contribution is used to calculate the **break-even point**



Worked Example

Rosebud Aromas manufactures luxury scented candles. The production of each candle incurs the following VC:

VC	Cost per candle (£)
Wax	0.14
Perfume oil	0.72
Glass jar	1.46
Outer packaging	0.33

Each candle is sold for an average wholesale price of £15 to retail outlets. Calculate the contribution for each candle.

Step 1: Calculate the total VC per candle

$$= £0.14 + £0.72 + £1.46 + £0.33$$

$$= £2.65$$

Step 2: Deduct the total VC per candle from the selling price

$$= £15.00 - £2.65$$

$$= £12.35$$



Your notes



Using the contribution to calculate the break-even point

- The contribution value is used to calculate the break-even point
- The break-even point is where **total revenue earned for a product is exactly equal to its total costs** and where the business is making **neither a profit nor a loss**
- It is calculated using the formula

$$\text{Break - even point} = \frac{\text{Fixed costs}}{\text{Contribution}}$$

- The break-even point is expressed in **units** (e.g. the number of scented candles)
- Identifying the break-even point allows a business to understand **how many items it needs to produce and sell** to cover all costs before it starts to make a profit
 - Each subsequent unit sold past this point will generate profit for the business



Worked Example

Cost and revenue data for Canterbury Glamping

Cost/revenue	Cost (£)
Revenue per pod per night	95
Variable costs per pod per night	19
Annual fixed costs	55,000

Using the information in the table, calculate how many pods need to be occupied each month for Canterbury Glamping to break even. [4]

Step 1: State the formula to calculate the break-even point

$$\text{Break - even point} = \frac{\text{Fixed costs}}{\text{Contribution}} \quad [1]$$

Step 2: Calculate the contribution



Your notes

$$= \text{Selling price} - \text{variable cost per unit}$$

$$= £95 - £19 \quad [1]$$

$$= £76$$

Step 3: Apply the formula to calculate the break-even point

$$= \frac{£55,000}{£76} \quad [1]$$

$$= 723.68$$

Step 4: Always round up to the nearest whole number because only whole products can be sold

$$= 724 \text{ camping pods } [1]$$

Margin of safety

- The margin of safety is the **difference** between the actual level of output of a business and its **break-even level of output**
- The margin of safety can be calculated using the following formula:

$$\text{Margin of safety} = \text{Actual level of output} - \text{Break - even level of output}$$



Worked Example

The cost, sales and revenue for an electric bicycle manufacturer are presented in the table below.

Annual fixed costs	£42,000
Selling price per unit	£750
Variable cost per unit	£350
Number of units sold	240

Using the data, calculate the margin of safety. You are advised to show your workings.

Step 1: Calculate the contribution

$$= £750 - £350$$

$$= £400$$

Step 2: Calculate the break-even point

$$\begin{aligned} &= \frac{\pounds 42,000}{\pounds 400} \\ &= 105 \text{ units} \end{aligned}$$

Step 3: Calculate the margin of safety

$$\begin{aligned} &= 240 \text{ units} - 105 \text{ units} \\ &= 135 \text{ units} \end{aligned}$$

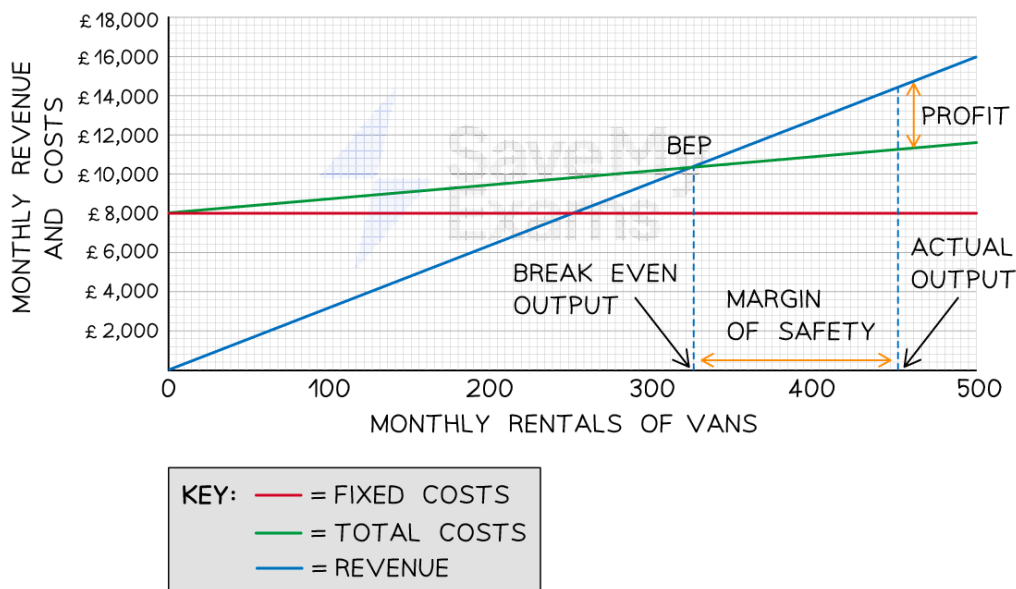


Your notes

Interpretation of break-even charts

- A break-even chart is a **visual representation** of the break-even point and is used to identify the following
 - **Fixed costs, total costs and revenue** over a range of output
 - The **break-even point**, where total costs are equal to revenue
 - The **profit or loss** made at each level of output
 - The **margin of safety**

An example break-even chart



The break-even chart for A2B Limited shows that at 324 units, the total revenue = the total costs

Diagram analysis



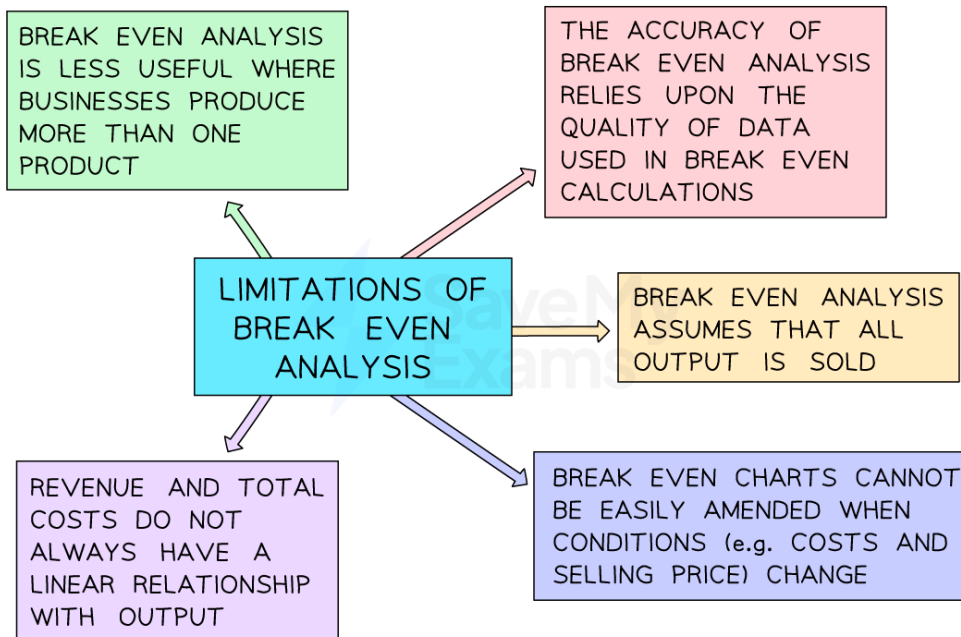
Your notes

- Fixed costs do not change as output increases
 - A2B's fixed costs are £8,000, and these do not change whether the business produces zero units or 500 units
- Total costs are made up of fixed and variable costs
 - At zero units of output, they are made up exclusively of fixed costs
 - At 500 units, the total variable costs equate to £11,800
 - This line **slopes upwards** because total variable costs increase as output increases
- The revenue line also slopes upwards
 - At zero units of output, the revenue is £0
 - At 500 units, the total revenue equates to £11,800
 - Revenue will increase with the output
 - The line will **slope more steeply than the total costs** and will **cross the total costs line** at some point
- The point at which the total costs and the revenue lines cross is the **break-even point**
 - The break-even level of output for A2B is 324 units
- **The margin of safety can be identified as the difference on the x-axis between the actual level of output (in this case, 450 units) and the break-even point**
- The profit made at a specific level of output can be identified as the **space between the revenue and total cost lines**
 - In this instance, the profit made at 450 units of output is $£14,400 - £11,250 = £3,150$

Limitations of break-even analysis



Your notes



The limitations of break-even analysis include the assumption that all output is sold



Examiner Tips and Tricks

When assessing break-even analysis, ensure that you explain why it has an important internal planning role, but don't forget that it has a significant external role as well.

Break-even analysis should be included in a business plan when a business is trying to secure external finance. Businesses looking to borrow money or attract investors seeking to manage their risk should take care to model the break-even point, margin of safety and level of profit (or loss) at different levels of output and prepare themselves to be scrutinised on the figures.



The purpose of budgets

- A budget is a **financial plan** that a business (or department in the business) sets for costs and revenue
- The budget is usually closely aligned with the **business objective**

Reasons for using budgets

Reason	Explanation
Planning and monitoring	▪ Businesses that use budgets are actively planning ahead ▪ Problems and their solutions may be considered and solved in advance
Control	▪ Frequent monitoring of budgets allows managers to precisely control their functional area ▪ Budgets support the setting and review of company or department objectives
Coordination and communication	▪ Budgeting requires different parts of a business to operate as part of a coordinated whole ▪ Budgets may be communicated throughout the organisation to provide a framework for decision-making and communication
Motivation and efficiency	▪ Budgets play an important role in target-setting and performance management, which can be used by managers to measure success ▪ The allocation of budgets spreads decision-making across the organisation, acting as a motivator to the managers who control the budgets

Types of budgets

- Budgets are usually **set annually** and then monitored on a monthly basis
- Businesses may set budgets to monitor the financial performance of any aspect of the business
- Budgets are generally prepared using one of two methods:
 - Historical figure budgets

- Zero-based budgeting

Historical figure budgets

- Budgets are usually **based on historical data** (e.g. sales and costs data from previous years) and allow for factors such as **inflation** and other relevant economic indicators (e.g. **exchange rate** variations)

Zero-based budgeting

- In some cases, businesses make the decision **not to allocate budgets** and use a **zero-budgeting** approach
 - This is particularly useful when a business needs to **control costs closely** (e.g. to improve profitability)
- Zero budgeting requires **all spending to be justified**, which means that many unnecessary costs can be eliminated
 - It can be **time-consuming**, as evidence to support spending decisions needs to be collected and presented
 - Zero budgeting also **requires skilled and confident employees** to make a persuasive case to convince those who are making purchasing decisions

Variance analysis

- A **budget variance** is the **difference between the figure budgeted and the actual figure achieved** by the end of the budgetary period (e.g. 12 months)
- **Variance analysis** seeks to determine the reasons for the differences between the actual figures and the budgeted figures

Favourable variances

- A **favourable variance** is when the **actual figure achieved is better than the budgeted figure**
 - A favourable variance in a **revenue or profit budget** is when the actual figure is **higher than the budgeted figure**
 - A favourable variance in a **costs budget** is when the actual figure is **lower than the budgeted figure**

Adverse variances

- An **adverse variance** is when the **actual figure achieved is worse than the budgeted figure**
 - An adverse variance in a revenue or profit budget is when the actual figure is **lower** than the budgeted figure
 - An adverse variance in a costs budget is when the actual figure is **higher** than the budgeted figure



Your notes





Worked Example

Selected financial information for Bunsen PLC in 2022

	£ m
Budgeted sales revenue	12,460
Actual sales revenue	13,718
Budgeted total costs	8,420
Actual total costs	10,627

Using the data, calculate the total profit variance for Bunsen PLC in 2022. You are advised to show your workings. [4]

Step 1: Calculate the budgeted profit for 2022

$$\begin{aligned} &= £12,460 - £8,420 \\ &\quad [1] \\ &= £4,040 \end{aligned}$$

Step 2: Calculate the actual profit for 2022

$$\begin{aligned} &= £13,718 - £10,627 \\ &\quad [1] \\ &= £3,091 \end{aligned}$$

Step 3: Subtract the budgeted profit from the actual profit for 2022

$$\begin{aligned} &= £3,091 - £4,040 \\ &\quad [1] \\ &= £949 \end{aligned}$$

Step 4: Identify the nature of the variance

- In this case, the variance is adverse because the actual profit for 2022 is lower than the budgeted profit for 2022
- The correct answer is £949 in adverse variance [1]



Examiner Tips and Tricks

Although the Bunsen PLC example shows an adverse profit variance, it is worth noting that the company's actual sales revenue was higher than budgeted. There could be some sales executives in the business who deserve some sincere congratulations!

You may recommend that the business investigate the reasons for the adverse profit variance. The focus of the examination must be on the higher-than-budgeted costs



Your notes

rather than on this seemingly positive sales performance. You may recommend that Bunsen review its supply agreements or that it adopt a zero-budgeting approach.



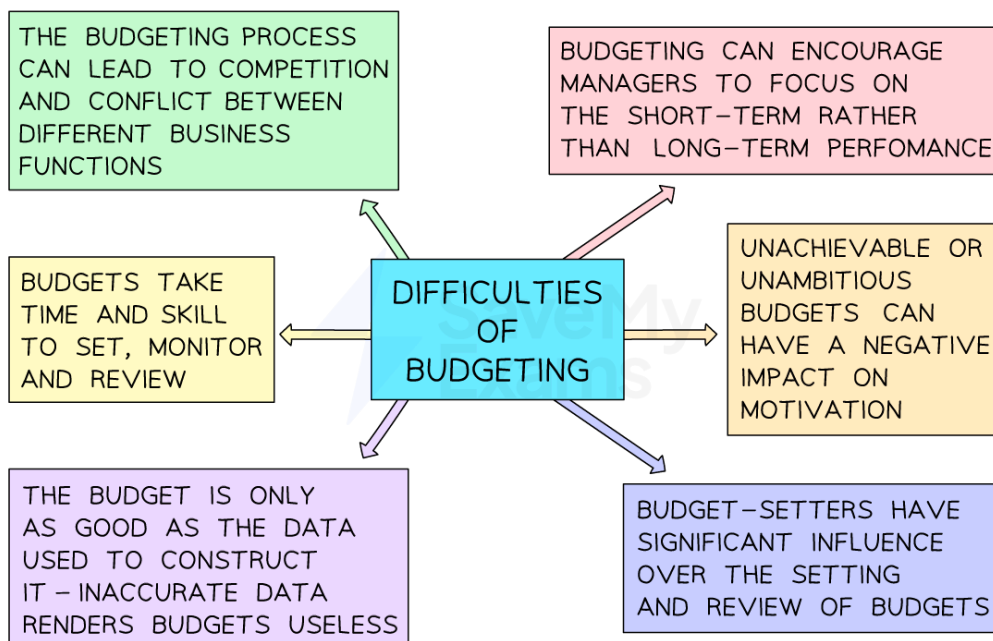
Your notes

- Once variances have been identified, a business should **carefully investigate the reasons why the variances have occurred** and take appropriate action, such as:
 - Where adverse cost variances are identified, a business may **seek alternative suppliers** or investigate ways to improve efficiency
 - Where adverse sales variances are identified, a business may **review its marketing activities** to improve their effectiveness
 - Where favourable cost variances are identified, a business may review key quality indicators such as the volume of returns or waste levels to **ensure that output standards are being met**
 - Where favourable sales variances occur, a business may **reward client-facing staff** with performance-based incentives

Difficulties of budgeting

- Budgeting requires **significant expertise** to be of genuine use to a business, and there are several difficulties associated with its construction

The main difficulties of budgeting



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Difficulties of budgeting include the time and skill to set, monitor and review them, as well as unachievable budgets having a negative impact on motivation

- Data must be **up-to-date, accurate and free of bias**
 - Sources of data must be selected carefully and used with care to ensure the most appropriate assumptions are made



Your notes

- **Those who are constructing budgets will require skills and relevant experience to do so effectively**
 - This may involve training or the recruitment of specialist staff
- Budgets can encourage managers to focus on the **short-term rather than the long-term success** of the business, as budgets are usually set year to year
 - **Unrealistic budgets** (over or under) can lead to a **lack of motivation** among those tasked to achieve the financial plan
- **Conflict between budget holders** may arise, reducing the effectiveness of the business as a whole